

Software Requirements Specification (SRS)

Government Payment Analytics Portal (MVP)

Version: 1.0

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1. Objective

The internship team has already developed a working Government Payment Analytics solution using Python, machine learning, and Streamlit. The objective of this project is to convert the existing prototype into a professional web application that government jurisdictions can access through a browser. This phase focuses on reusing the existing analytics engine and dashboard logic. No new analytics or machine learning functionality is required.

2. Existing Components

- Python analytics engine
- Payment risk prediction model
- Working capital calculations
- Feature importance analysis
- Executive dashboard
- Government payment dataset
- Dashboard visualizations

The goal is to integrate these components into a production-style web application.

3. Scope

The MVP should include:

- Secure Login Page
- Dashboard Home Page
- Upload Dataset Page
- Analytics Dashboard
- Report Export (optional if time permits)

Anything outside this scope should be considered Phase 2.

4. Application Pages

Login

Simple email/password login.

Dashboard Home

Display KPI cards: Total Transactions, Average Processing Lag, Average Risk, Total Obligations, Estimated Cashflow Penalty.

Upload Dataset

Allow Excel upload. Once uploaded, run the existing Python analytics engine and populate the dashboard.

Analytics Dashboard

Recreate the current Streamlit dashboard with Executive Summary, County Performance, High Risk Transactions, Feature Importance, and Working Capital Impact. Reuse the existing calculations and charts wherever possible.

5. Navigation

Login → Dashboard → Upload Dataset → Analytics Dashboard → Logout

6. Technology Stack

Frontend: React, Next.js, Tailwind CSS

Backend: FastAPI (Python)

Analytics: Existing Python codebase

Database: Not required for MVP (file-based processing).

7. Deliverables

- Responsive web application
- Login page
- Upload page
- Analytics dashboard
- Integration of the existing analytics engine
- Existing charts and visualizations

8. Out of Scope

ERP integrations, APIs, database implementation, multi-tenant support, AI chatbot, email alerts, additional analytics models, and administrative portal.

9. Success Criteria

Users can log in, upload a payment dataset, have the existing analytics engine process the file, view the same analytics and charts currently available in the Streamlit prototype, and deploy the application as a standard web application.